Name: _____ Period: _____ Date: _____

Topic 4 Practice Test: 4.0, 4.2–4.6

Show all your work. Indicate clearly the methods you use, because you will be graded on the correctness of your methods as well as the accuracy of your final answers. No calculator.

Multiple Choice (#1–5). Identify the choice that best completes the statement or answers the question.

1. _____ If $\Delta MNP \cong \Delta QRS$, then $\overline{PM} \cong$

A $\overline{QR}$ **B** $\overline{RS}$ **C** $\overline{SQ}$ **D** $\overline{NP}$

2. _____ Is the following statement true: A right triangle can also be an obtuse triangle.

A Always True **B** Sometimes True **C** Never True **D** Can't be determined

3. _____ Which is a correct classification of ΔABC if $m\angle A = 38°$ and $m\angle B = 52°$?

A Acute **B** Right **C** Obtuse **D** Equiangular

4. _____ Is the following statement true: An isosceles triangle is also an equilateral triangle.

A Always True **B** Sometimes True **C** Never True **D** Can't be determined

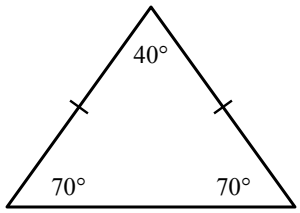
5. _____ Which of the following can <u>not</u> be used to prove two triangles are congruent?

A SAS **B** AAS **C** SSA **D** HL

Classifying Triangles (#6–11).

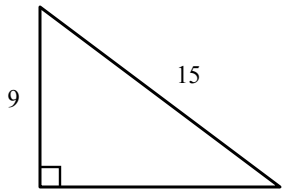
6. Classify each triangle by its angles *and* sides.

a.



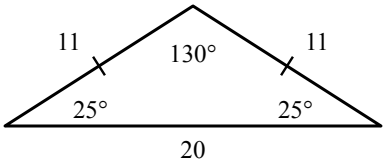
_____

b.



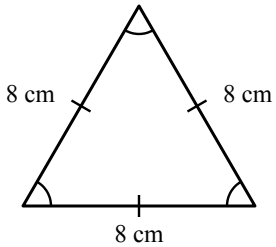
_____

c.



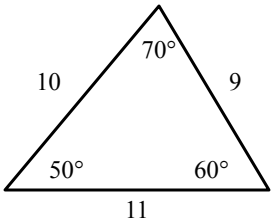
_____

d.



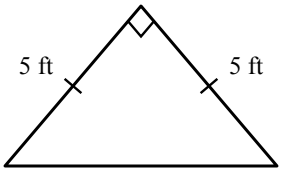
_____

e.



_____

f.



_____

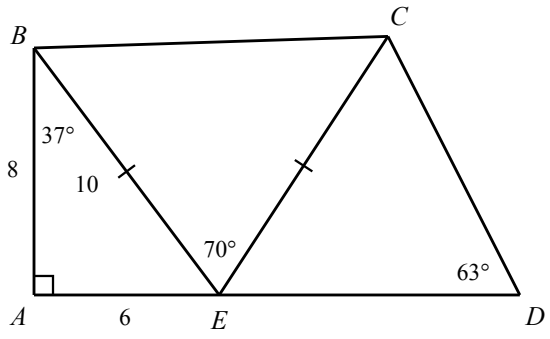
7. Points A, E, and D are on one line. Classify each triangle by its angles and sides. *(Find the missing angles first.)*

a. ΔABE: _____

b. ΔBEC: _____

c. ΔCED: _____

d. $m\angle ECD =$ _____



8. Use the distance formula to find each side length, then classify.

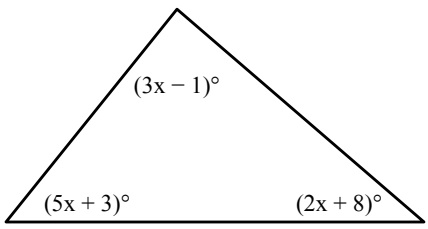
a. $A(1, 2)$, $B(4, 6)$, $C(8, 3)$. Classify by sides and angles.

b. $P(0, 0)$, $Q(6, 0)$, $R(2, 5)$. Classify by sides.

9. In ΔABC, the measure of $\angle B$ is 12 more than twice the measure of $\angle A$, and the measure of $\angle C$ is 12 less than three times the measure of $\angle A$. Find the measure of each angle, then classify ΔABC by its angles and sides.

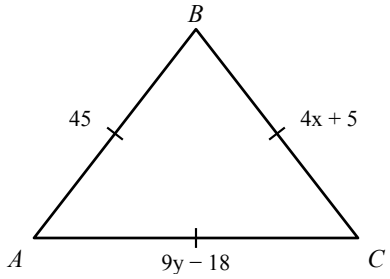
10. In a right triangle, the measure of one acute angle is 18° less than twice the measure of the other acute angle. Find the measures of the acute angles.

11. Find the value of x. Then classify the triangle by its angles and sides.



Isosceles and Equilateral Triangles (#12–21).

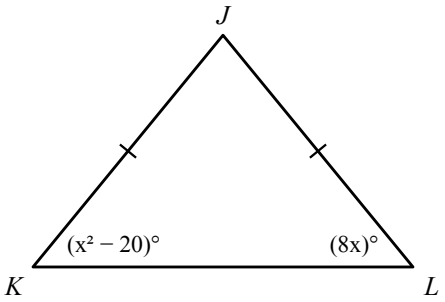
12. If ΔABC is an equilateral triangle, solve for both x and y.



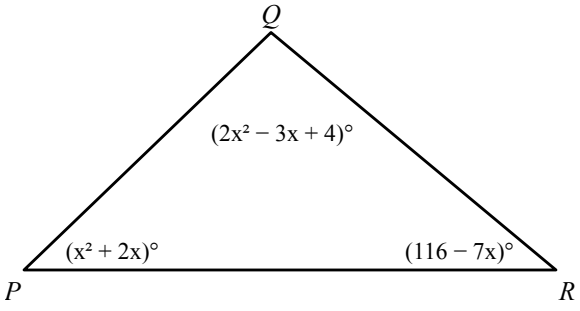
13. ΔSTU is an equilateral triangle. If ST is 3 less than twice x, TU is 6 more than x, and SU is 21 less than four times x, find x and the measure of each side.

14. ΔKLM is an isosceles triangle with $\overline{KL} \cong \overline{KM}$. If $KL = 5x - 8$, $KM = 3x + 6$, and $LM = 2x + 3$, find x, the measure of each side, and the perimeter.

15. In ΔJKL, $\overline{JK} \cong \overline{JL}$. Solve for x. Then find $m\angle J$ and classify ΔJKL by its angles and sides.



16. Using the integer value of x, classify ΔPQR by its angles and sides.



17. Find and use the value of x that makes sense to find FI and IN. Then classify ΔFIN by its angles and sides.

$FI = (2x - 3)^2$
$IN = 6(3 - 2x)$
$FN = 36$

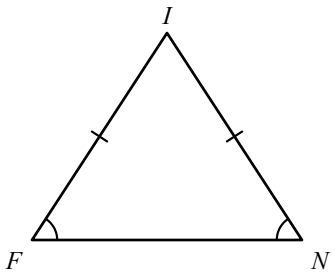
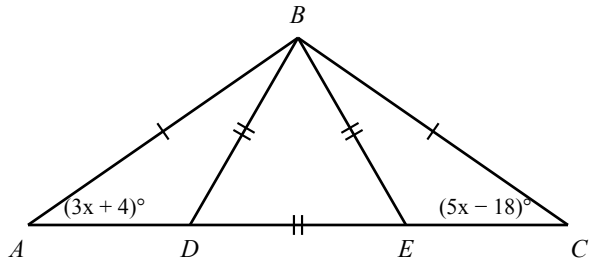


Figure not drawn to scale.

18. If ΔABC is isosceles with $\overline{AB} \cong \overline{CB}$ and ΔDBE is an equilateral triangle, find each value. Points A, D, E, C are on one line.



$x =$ _____

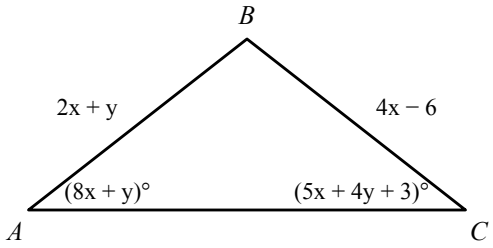
$m\angle A =$ _____

$m\angle ADB =$ _____

$m\angle ABD =$ _____

$m\angle ABC =$ _____

19. ΔABC is isosceles with vertex B. Solve for x and y. Then find $m\angle B$ and classify ΔABC by its angles. *Hint: you will need a system of equations.*



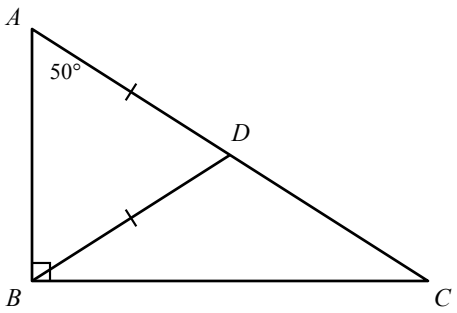
20. In right ΔABC, D is on $\overline{AC}$ and $\overline{AD} \cong \overline{BD}$. If $m\angle A = 50°$, find each measure and classify ΔBDC.

a. $m\angle ABD =$ _____

b. $m\angle DBC =$ _____

c. $m\angle BDC =$ _____

d. ΔBDC: _____



21. ERROR ANALYSIS In ΔXYZ, $\overline{XY} \cong \overline{XZ}$ and $m\angle X = 40°$. Describe and correct the error below.

```
Base angles are congruent, so
m∠Y = m∠X = 40°
m∠Z = 180 − 40 − 40 = 100°
```

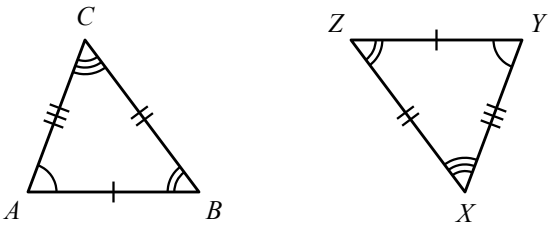
Congruence Statements and CPCTC (#22–31).

22. Write three valid congruency statements for the triangles below.

a. _____

b. _____

c. _____



23. Given $\Delta JKL \cong \Delta PQR$, complete each of the following statements.

a) $\angle K \cong$ _____ b) $\overline{LJ} \cong$ _____ c) $\Delta LKJ \cong$ _____

d) _____ $\cong \overline{QR}$ e) $\angle R \cong$ _____ f) $\Delta QPR \cong$ _____

24. Given $\Delta DGK \cong \Delta RTW$, list all congruent angles and sides.

Angles	Sides

25. $\Delta FGH \cong \Delta LMN$. $FG = 12$ cm, $GH = 9$ cm, $NL = 17$ cm, $m\angle F = 34°$, and $m\angle H = 105°$. Find each missing measure. *Hint: draw the triangles if needed.*

a) $LM =$ _____ b) $MN =$ _____ c) $FH =$ _____

d) $m\angle L =$ _____ e) $m\angle N =$ _____ f) $m\angle M =$ _____

26. $\Delta ABC \cong \Delta DEF$. $AB = 3x + 4$, $DE = 5x - 10$, $BC = x + y$, $m\angle C = (4y + 6)°$, and $m\angle F = 70°$. Find x, y, and EF.

27. $\Delta XYZ \cong \Delta UVW$. $m\angle X = 47°$, $m\angle W = 90°$, $m\angle Y = (6y + 1)°$, $m\angle V = (4z - 1)°$, $XZ = 3x - 5$, and $UW = 2x + 9$. Find x, y, and z.

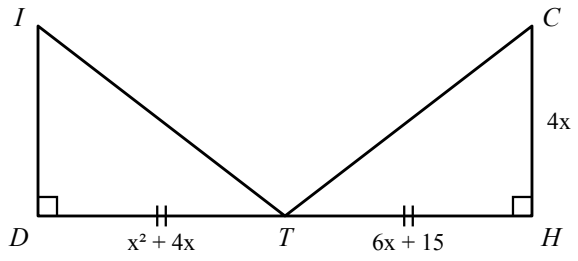
28. $\Delta JKL \cong \Delta MNO$. Solve for x and y, then find NO.

$JK = 3x + y$, $JL = 4x - y$, $KL = 2x + 3y$, $MN = 18$, $MO = 17$

29. $\Delta ABC \cong \Delta XYZ$. $AB = 2x - 4$, $BC = x + 9$, $AC = 3x - 10$, and $XY = 18$. Find the perimeter of ΔABC.

30. $\Delta GHI \cong \Delta MNO$. $GH = x^2 - 3x$, $MN = 2x + 14$, and $GI = 5x + 6$. Solve for x, then find GH. Check every solution.

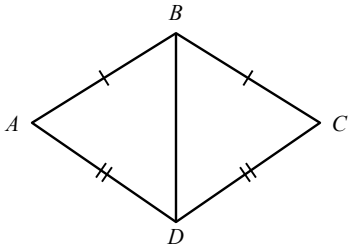
31. $\Delta TID \cong \Delta TCH$. Solve for x. Then find the area of ΔTID.



Triangle Congruence Shortcuts: SSS, SAS, ASA, AAS, HL (#32–42).

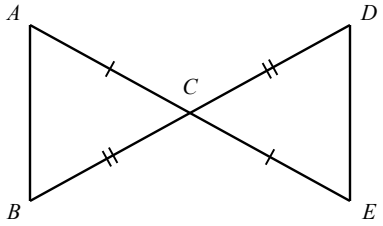
32. Complete the triangle congruence statement and tell which shortcut supports it. If the triangles cannot be shown to be congruent, write "CBD" (cannot be determined) and leave the statement blank.

a.



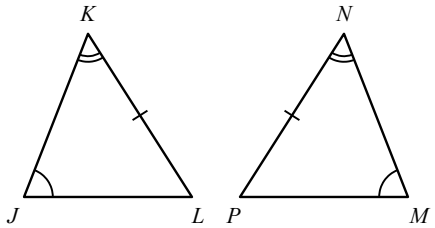
$\Delta ABD \cong$ _____ Reason: _____

b.



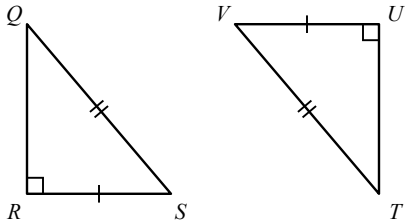
$\Delta ABC \cong$ _____ Reason: _____

c.



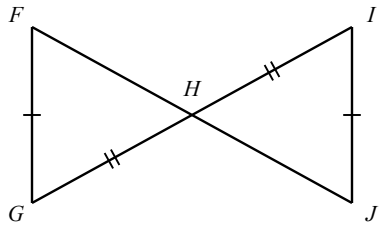
$\Delta JKL \cong$ _____ Reason: _____

d.



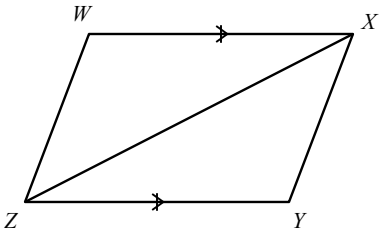
$\Delta QRS \cong$ _____ Reason: _____

e.



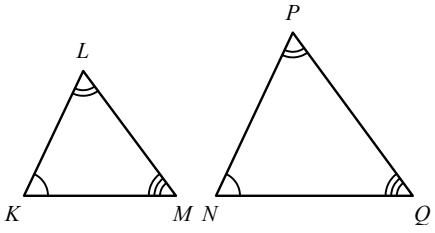
$\Delta FGH \cong$ _____ Reason: _____

f.



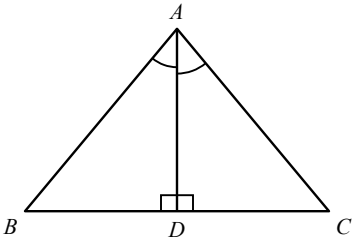
$\Delta WXZ \cong$ _____ Reason: _____

g.



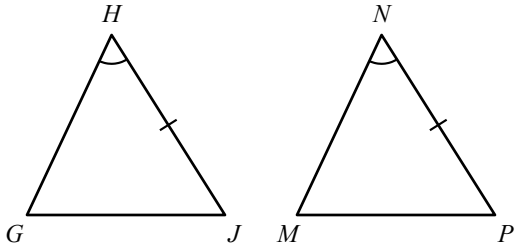
$\Delta KLM \cong$ _____ Reason: _____

h.



$\Delta ABD \cong$ _____ Reason: _____

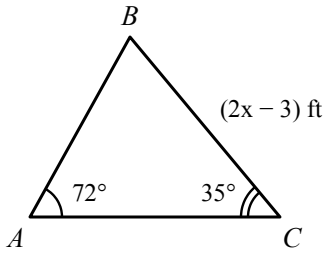
33. What additional piece of information is needed to show $\Delta GHJ \cong \Delta MNP$ by each shortcut?

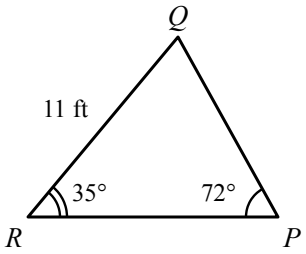


a) SAS: _____ b) ASA: _____ c) AAS: _____

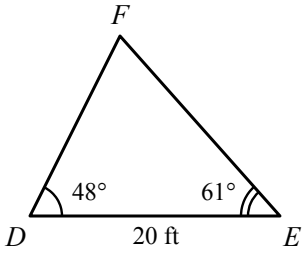
34. For what value of x would each pair of triangles be congruent by the given shortcut?

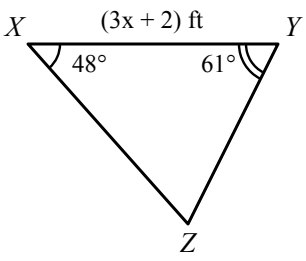
a. AAS



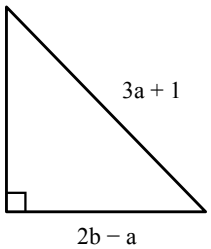


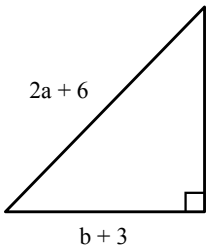
b. ASA



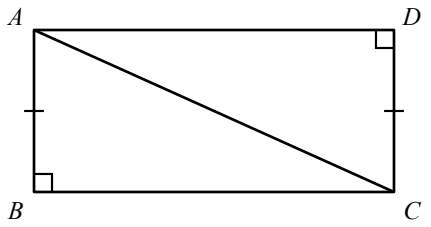


35. For what values of a and b are the triangles congruent by the HL Theorem? The labeled legs correspond.

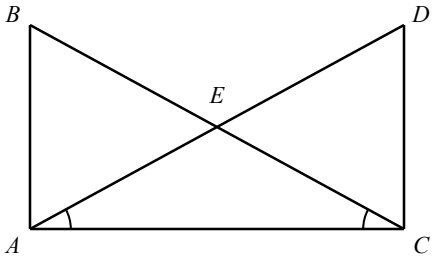




36. **Understand** Is there sufficient information to prove $\Delta ABC \cong \Delta CDA$ by SSS? Explain.



37. $\overline{AD}$ and $\overline{CB}$ intersect at E, and $\angle EAC \cong \angle ECA$.



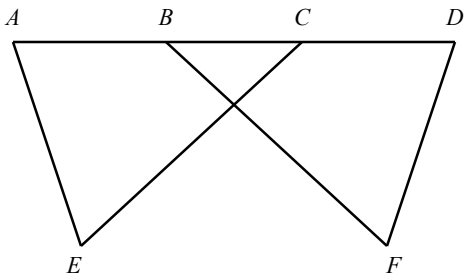
a. Why is $\overline{AE} \cong \overline{CE}$?

b. What additional statement do you need to prove $\Delta ABE \cong \Delta CDE$ by SAS? _____

c. What additional statement do you need to prove $\Delta ABE \cong \Delta CDE$ by AAS? _____

d. Suppose $\Delta ABE \cong \Delta CDE$. Explain why $\Delta ABC \cong \Delta CDA$ and name the shortcut.

38. Points A, B, C, D are on one line and $\Delta ACE \cong \Delta DBF$. Identify which side or angle is congruent to each given part.



a. $\angle ACE \cong$ _____

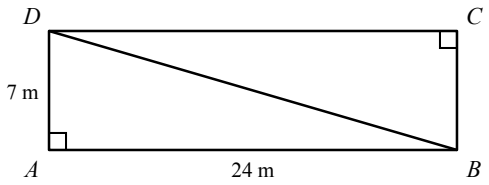
b. $\overline{CE} \cong$ _____

c. $\angle AEC \cong$ _____

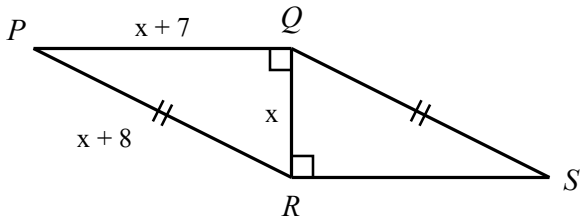
d. Explain why $\overline{AB} \cong \overline{CD}$.

39. **Apply** A triangular piece for a phone stand has sides of 9 cm and 12 cm with a 50° angle between them. The slot it goes into is a triangle with sides of 12 cm and 9 cm with a 50° angle between them. Will the piece fit exactly in the slot? Explain.

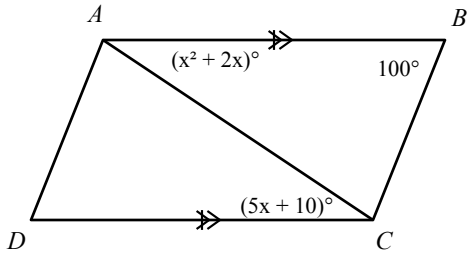
40. A drone flies the path $A \to B \to D \to C$ at a constant speed of 4 meters per second. In the figure, $\Delta ABD \cong \Delta CDB$. How long does the trip take? Round to the nearest tenth of a second.



41. Solve for x. Explain why $\Delta PQR \cong \Delta SRQ$, then find the perimeter of ΔSRQ.



42. $\overline{AB} \parallel \overline{DC}$ and $\overline{AB} \cong \overline{DC}$. Solve for x, then find each measure.



a. $x =$ _____

b. $m\angle BAC =$ _____

c. $m\angle BCA =$ _____

d. $\Delta ABC \cong \Delta CDA$ by _____

e. $m\angle D =$ _____

Challenge (#43–45). These take several steps. Check every solution you get from factoring.

43. Solve for x and y. Then find all three angle measures and classify the triangle by its angles and sides.

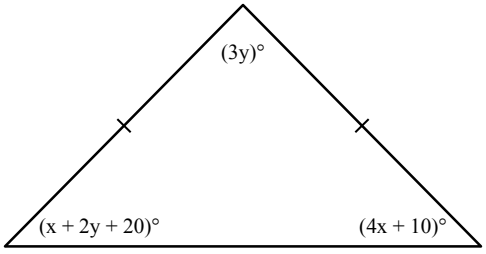


Figure not drawn to scale.

44. ΔABC has vertices $A(-4, 1)$, $B(-1, 5)$, $C(-1, 1)$. ΔDEF has vertices $D(2, -1)$, $E(2, -4)$, $F(6, -4)$.

a. Prove the triangles are congruent. Write a congruence statement with the vertices in the correct order and name the shortcut.

b. Find $m\angle A + m\angle F$ without a protractor. Explain.

c. Classify ΔDEF by its angles and sides, and find its area.

45. $\Delta ACB \cong \Delta DCE$. Points A, C, D are on one line, and points B, C, E are on one line. Find x and y. Then find $m\angle B$ and classify ΔDCE by its angles and sides.

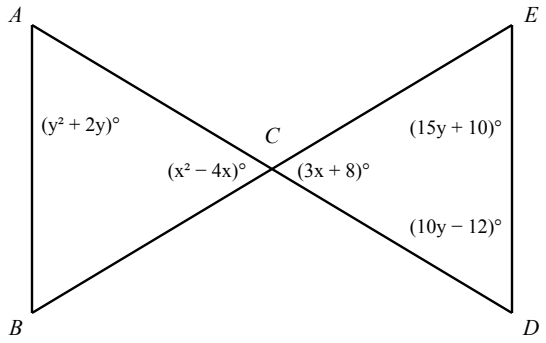


Figure not drawn to scale.